

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	9 th march 2026
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	4 Marks

Technical Architecture:

Tech Stack Overview

Frontend (User Interface)



Backend (Application Logic)



Message Broker



Machine Learning



Database



External APIs



Deployment & Infrastructure



AI-Driven Healthcare Anomaly Detection System Architecture

- 1 **Patient Sensors / Data Generator**
- Heart Rate, SpO₂, Temperature, Blood Pressure
- HTML5 CSS3 JS Jinja2

Real-Time Data Streams

- 2 **Apache Kafka**
(WijPrfventor)



Anomaly Scores & Alerts

- 3 **ML Inference Engine** (Autoencoder + Isolation Forest)



Anomaly Score

LOW MEDIUM HIGH



Logs Historical Data

- 4 **Flask Backend**



Anomaly Scores & Alerts



Web Dashboard

- 5 **Data Storage & Backend Logic**



PostgreSQL



Email Alerts (SMTP)

- 5 **Layer 1: Data Ingestion** (Sensors)

- 4 **Layer 2: Stream Processing**

- 3 **Layer 3: Machine Learning Inference Engine** (Autoencoder + Isolation Forest)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive web dashboard for nurses to view real-time patient vitals and anomaly alerts.	HTML, CSS, JavaScript, Flask Templates
2.	Application Logic-1	Real-time data streaming pipeline to continuously ingest patient vital signs without data loss.	Apache Kafka, Python
3.	Application Logic-2	Backend routing, database interactions, and integration of the ML inference engine.	Python, Flask Web Framework
4.	Application Logic-3	Alerting mechanism logic to trigger actions based on LOW, MEDIUM, or HIGH severity levels.	Python
5.	Database	Relational database to store patient historical vitals, anomaly scores, and alert logs safely.	PostgreSQL
6.	Cloud Database	Managed database service for production environments (if deployed to the cloud).	Amazon RDS for PostgreSQL (or local DB)
7.	File Storage	Storing trained ML model weight files (.h5 or .pkl) and application system logs.	Local Filesystem / Docker Volumes

S.No	Component	Description	Technology
8.	External API-1	Email delivery service to automatically notify healthcare admins of HIGH severity events.	SMTP / SendGrid API
9.	External API-2	Simulating patient physiological data (since real patient hardware isn't connected).	Custom Data Generator API (Python)
10.	Machine Learning Model	Unsupervised models to learn normal patient baselines and detect sudden physiological outliers.	Autoencoder (TensorFlow/Keras) & Isolation Forest (Scikit-learn)
11.	Infrastructure (Server / Cloud)	Application Deployment environment hosting the Flask app, Kafka broker, and ML scripts.	Docker Containers running on AWS EC2 or Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Core libraries utilized for streaming, backend routing, and machine learning.	Apache Kafka, Flask, Scikit-learn, TensorFlow

S.No	Characteristics	Description	Technology
2.	Security Implementations	Securing sensitive patient data, protecting database credentials, and securing email transmissions.	Environment Variables (.env), basic Flask-Login authentication, TLS/SSL for SMTP
3.	Scalable Architecture	Event-driven and decoupled design. Kafka separates data ingestion from ML processing, allowing it to handle thousands of patients.	Publish-Subscribe Architecture (Kafka) & Containerization (Docker)
4.	Availability	Ensuring the continuous monitoring system stays online and recovers quickly from crashes.	Docker Container restart policies, persistent DB volumes
5.	Performance	Guaranteeing low-latency (< 2 seconds) anomaly detection and real-time dashboard UI updates without page reloads.	WebSockets or asynchronous AJAX, Kafka low-latency messaging

References:

[\[1\] A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, 2019.](#)

[\[2\] F. T. Liu, K. M. Ting, and Z. Zhou, "Isolation Forest," IEEE ICDM, 2008.](#)

[\[3\] Apache Kafka Documentation. <https://kafka.apache.org>](#)